

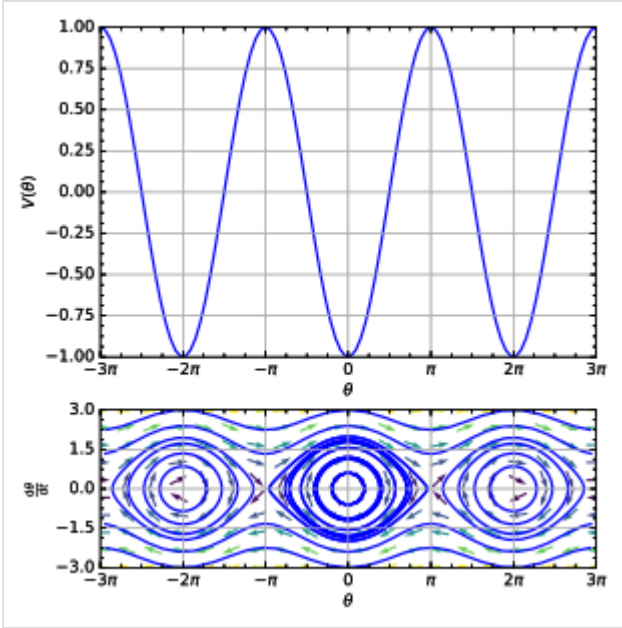


Phase portrait

In mathematics, a **phase portrait** is a geometric representation of the orbits of a dynamical system in the phase plane. Each set of initial conditions is represented by a different point or curve.

Phase portraits are an invaluable tool in studying dynamical systems. They consist of a plot of typical trajectories in the phase space. This reveals information such as whether an attractor, a repellor or limit cycle is present for the chosen parameter value. The concept of topological equivalence is important in classifying the behaviour of systems by specifying when two different phase portraits represent the same qualitative dynamic behavior. An attractor is a stable point which is also called a "sink". The repeller is considered as an unstable point, which is also known as a "source".

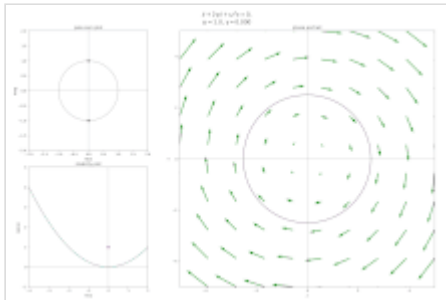
A phase portrait graph of a dynamical system depicts the system's trajectories (with arrows) and stable steady states (with dots) and unstable steady states (with circles) in a phase space. The axes are of state variables.



Potential energy and phase portrait of a simple pendulum. Note that the x-axis, being angular, wraps onto itself after every 2π radians.

Examples

- Simple pendulum, see picture (right).
- Simple harmonic oscillator where the phase portrait is made up of ellipses centred at the origin, which is a fixed point.
- Damped harmonic motion, see animation (right).
- Van der Pol oscillator see picture (bottom right).



Phase portrait of damped oscillator, with increasing damping strength. The equation of motion is $\ddot{x} + 2\gamma\dot{x} + \omega^2x = 0$.

Visualizing the behavior of ordinary differential equations

A phase portrait represents the directional behavior of a system of ordinary differential equations (ODEs). The phase portrait can indicate the stability of the system. ^[1]

Stability ^[1]	
Unstable	Most of the system's solutions tend towards ∞ over time
Asymptotically stable	All of the system's solutions tend to 0 over time
Neutrally stable	None of the system's solutions tend towards ∞ over time, but most solutions do not tend towards 0 either

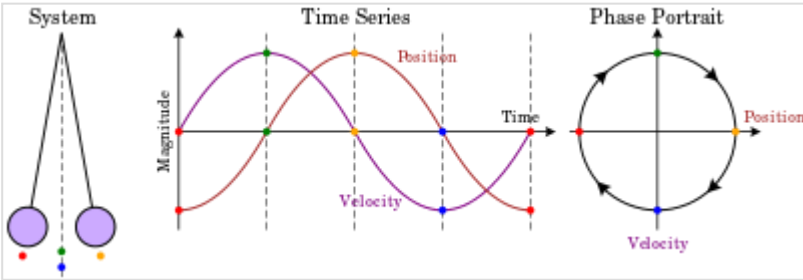


Illustration of how a phase portrait would be constructed for the motion of a simple pendulum.

The phase portrait behavior of a system of ODEs can be determined by the eigenvalues or the trace and determinant (trace = $\lambda_1 + \lambda_2$, determinant = $\lambda_1 \times \lambda_2$) of the system. ^[1]

Eigenvalue, Trace, Determinant	Phase Portrait Shape
λ_1 & λ_2 are real and of opposite sign; Determinant < 0	Saddle (unstable)
λ_1 & λ_2 are real and of the same sign, and $\lambda_1 \neq \lambda_2$; $0 < \text{determinant} < (\text{trace}^2 / 4)$	Node (stable if trace < 0, unstable if trace > 0)
λ_1 & λ_2 have both a real and imaginary component; $(\text{trace}^2 / 4) < \text{determinant}$	Spiral (stable if trace < 0, unstable if trace > 0)

See also

- Phase space
- Phase plane

References

1. Haynes Miller, and Arthur Mattuck. *18.03 Differential Equations*. Spring 2010. Massachusetts Institute of Technology: MIT OpenCourseWare, <https://ocw.mit.edu>. License: Creative Commons BY-NC-SA. (Supplementary Notes 26 by Haynes Miller: https://ocw.mit.edu/courses/18-03-differential-equations-spring-2010/resources/mit18_03s10_chapter_26/)

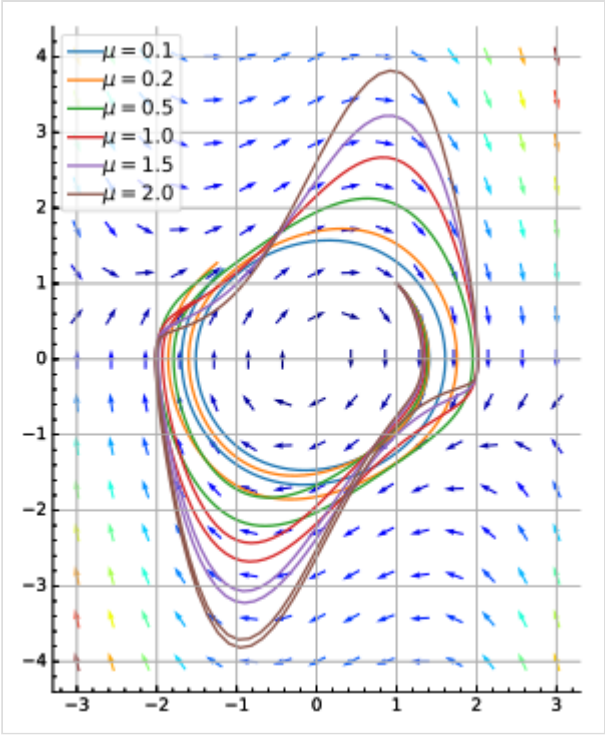
- Jordan, D. W.; Smith, P. (2007). *Nonlinear Ordinary Differential Equations* (fourth ed.). Oxford University Press. ISBN 978-0-19-920824-1. Chapter 1.
- Steven Strogatz (2001). *Non-linear Dynamics and Chaos: With applications to Physics, Biology, Chemistry and Engineering* (<https://archive.org/details/nonlineardynamic00stro>). ISBN 9780738204536.

External links

- Linear Phase Portraits (<http://mathlets.org/mathlets/linear-phase-portraits-matrix-entry/>), an MIT Mathlet.

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Phase portrait of van der Pol's equation,

$$\frac{d^2y}{dt^2} + \mu(y^2 - 1)\frac{dy}{dt} + y = 0.$$